Approximation and Rounding - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Round 6.847 to 1 decimal place.

The 2nd decimal digit is 4, so round down

Answer: 6.8

Question 2 *Practice question written for this worksheet (not from a past paper)*

Round 0.0583 to 2 significant figures.

First significant figure is 5; next digit is 3, round down

Answer: 0.058

Question 3 *Practice question written for this worksheet (not from a past paper)*

Round 45 719 to 3 significant figures.

457|19: next digit 1, round down; fill with zeros

Answer: 45 700

Question 4 *Practice question written for this worksheet (not from a past paper)*

Round 3.0962 to 2 decimal places.

The 3rd decimal digit is 6, so round up: 3.09 → 3.10

Answer: 3.10

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Round 0.0004096 to 3 significant figures.

Significant figures start at 4: 4, 0, 9|6

Round up: 4.10, so 0.000410

Answer: 0.000410

Question 6 *Practice question written for this worksheet (not from a past paper)*

Round 2 984 000 to 2 significant figures.

29|84…: next digit 8, round up to 30

Fill with zeros

Answer: 3 000 000

Question 7 *Practice question written for this worksheet (not from a past paper)*

A calculator shows 7.28571428.

(a) Write this to 3 significant figures.

(b) Write this to 1 decimal place.

(a) 3 s.f.: 7.28|57 → 7.29

(b) 1 d.p.: 7.2|857 → 7.3

Answer: (a) 7.29 (b) 7.3

Question 8 *Practice question written for this worksheet (not from a past paper)*

Estimate the value of $\frac{19.8 \times 4.1}{0.39}$

Round to 1 s.f.: $\frac{20 \times 4}{0.4} = \frac{80}{0.4}$

Answer: 200

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Approximation and Rounding - Mark scheme

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 2

Q	Working	Answer	Mark	Notes
	Any **two** from: 2 (or 4) **or** 60 **or** 9 (or 3)		2	M1 for rounding at least 2 of the 3 given numbers to one significant figure
	eg $\frac{2^2\times 60}{\sqrt{9}}=80$ or $\frac{4\times 60}{3}=80$	80 with rounded calculation seen		A1 dep on M1, all figures rounded correctly and 80 given. ignore any statements – just mark the rounded calculation
	Working required			
				Total 2 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2024 · Question 16

Q	Working	Answer	Mark	Notes
	30 or 50		2	M1
	Working required	1500		A1 cao dep on M1 1500 must come from correct figures of 30 **and** 50
				Total 2 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 6

6	$\frac{70\times 40}{0.02}$ or $\frac{68\times 40}{0.02}$ or $\frac{70\times 43}{0.02}$ or $\frac{68\times 43}{0.02}$		2	M1	for a correct expression using a suitable approximation. 0.02 is the only acceptable denominator.
	$\frac{70\times 40}{0.02}=140000$ **or** $\frac{68\times 40}{0.02}=\frac{2720}{0.02}=136000$ **or** $\frac{70\times 43}{0.02}=\frac{3010}{0.02}=150500$ **or** $\frac{68\times 43}{0.02}=\frac{2924}{0.02}=146200$	Correct figures		A1	If student says 'no' then do not award the A mark rounded expression and evaluated answer required Intermediate step required unless rounded to 1sf For each, ×50 (oe) may be seen in intermediate step. eg $\frac{68\times 40}{0.02}=2720\times 50=136000$
					Total 2 marks

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Approximation and Rounding - Mark scheme

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2025 · Question 19

Q	Working	Answer	Mark	Notes
	Any **two** from: 2 (or 4) **or** 60 **or** 9 (or 3)		2	M1 for rounding at least 2 of the 3 given numbers to one significant figure
	eg $\frac{2^2 \times 60}{\sqrt{9}} = 80 \text{ or } \frac{4 \times 60}{3} = 80$	80 with rounded calculation seen		A1 dep on M1, all figures rounded correctly and 80 given ignore any statements – just mark the rounded calculation
	Working required			
				Total 2 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 11

Q	Working	Answer	Mark	Notes
(a)	$0.48031(4...) + 0.45555(5...)$ **or** $\frac{61}{127} + \frac{41}{90}$		2	M1 Evaluate either fraction correctly as a decimal to at least 5 sf (rounded or truncated) or as a simplified fraction or an answer of 0.935, 0.936, 0.9358 or 0.9359
	Correct answer scores full marks (unless from obvious incorrect working)	0.93587(05162)		A1 Correct to at least 5 sf (rounded or truncated)
(b)		0.936	1	B1 ft if at least 4 sf given in (a)
				Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 11

Question		Working	Answer	Mark	Notes
	(a)	8.9285(71429) – 3.1622(7766)		2	M1 for 8.9285(71429) **or** 3.1622(776) **or** for an answer rounded or truncated to 2, 3 or 4 significant figures eg 5.7, 5.8, 5.76, 5.77, 5.766
		Correct answer scores full marks (unless from obvious incorrect working)	5.7662(93768)		A1 at least 4 decimal places rounded or truncated
	(b)		5.8	1	B1 ft from (a) provided 3 or more significant figures
					Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 11

Q	Working	Answer	Mark	Notes
(a)	eg 15.59 or 0.477 or 0.478 or 0.4778 or 0.4779 or $\frac{745}{1559}$		2	M1 for calculating the denominator or for answer with 3 or 4 dp or for the correct fraction
		0.47787(04298…)		A1 must have minimum of 5 dp
(b)		0.478	1	B1ft dep on at least 4 decimal places
				Total 3 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 1

1 (a)	2.14... ÷ 4.4	0.4874(456952)	2	M1	for 2.14.... or 4.4 or $\frac{\sqrt{115}}{22}$
				A1	Accept if first four sig figs correct
(b)		0.487	1	B1	ft if (a) > 3 sig figs
					Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 1

1. (a)	11.5 or 1.96 seen		2	M1	Also award for $5\frac{85}{98}$ or $\frac{575}{98}$ or answer of 5.9 or 5.87
		5.8673(46939...)		A1	for at least first 5 figures (ignore figures after the first five)
(b)		5.9	1	B1	ft from (a) if non-trivial
					Total 3 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2011 · Question 1

1. (a)	$\frac{24.1}{0.6} - 38.44 = 40.166... - 38.44$		2	M1	for 0.6 or $\frac{3}{5}$ or 40.166... (4 figures correct rounded or truncated) or $40\frac{1}{6}$ or 38.44 or $38\frac{11}{25}$
		1.726666667		A1	Accept if first 4 figures correct (rounded or truncated) Also accept 1.726 & or $\frac{259}{150}$ or $1\frac{109}{150}$
(b)		1.73	1	B1	ft from (a) if answer to (a) is a decimal with more than 3 sf
					Total 3 marks

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Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2023 · Question 14*

Q	Working	Answer	Mark	Notes
(a)	0.725(806...) or 1.53(9419...) or 14.8(4) or 2.26 or 2.27 or 2.265		2	M1
	Correct answer scores full marks (unless from obvious incorrect working)	2.2652(25539...)		A1
(b)		2.27	1	B1 ft from (a) dep on a number that has 4 or more significant figures
				Total 3 marks

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2013 · Question 2*

2	(a)	$\frac{2.720294102}{7.7}$		2	M1	for 2.72029... if first 5 figures correct (rounded or truncated) **or** for 7.7 **or** for $\frac{2\sqrt{185}}{77}$
			0.35328(4948)		A1	Accept if first 5 figures correct
	(b)		0.35	1	B1	ft from (a) only if more than 2 sig figs given in (a)
						Total 3 marks

CHALLENGE QUESTIONS

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 5*

5	(a)				M1	for $\frac{47}{32}$ **or** 1.46875 **or** $\frac{121}{25}$ **or** 4.84 **or** $\frac{5047}{800}$ **or** 6.30875 truncated or rounded to at least 1 dp
			6.30875	2	A1	
	(b)		6.31	1	B1	ft from (a) provided answer to (a) has more than 3 sig figs
						Total 3 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 17*

Question	Working	Answer	Mark	Notes
17	8.25, 8.35, 2.5, 1.5, 17.5, 12.5		3	B1 For any one of these stated or used accept $8.34\dot{9}$, $2.4\dot{9}$, $17.4\dot{9}$
	8.35(2.5+17.5) oe			M1 For $UB_a(UB_c+UB_y)$ $8.3 < UBa \leq 8.35$, $2 < UBc \leq 2.5$ $15 < UBy \leq 17.5$ (this allows for the student who uses some sort of upper value, but is slightly inaccurate, eg using 17.4 for y)
	Working required	167		A1 cao dep on previous marks
				Total 3 marks

Question 23 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 18*

18	7.45, 7.55, 3.415, 3.425, 1.5, 2.5		3	B1 For one correct upper or one correct lower bound Allow $7.54\dot{9}$ for 7.55, $3.424\dot{9}$ for 3.425, $2.4\dot{9}$ for 2.5
	$(X=)\dfrac{2\times7.55-3.415}{1.5}$ oe eg $\dfrac{11.685}{1.5}$			M1 $\dfrac{2\times UB_a - LB_b}{LB_f}$ where $7.5 < UB_a \le 7.55$, $3.415 \le LB_b < 3.42$, $1.5 \le LB_f < 2$ (also award this mark for $\dfrac{7.55-3.415}{1.5}$ or $\dfrac{2(7.55-3.415)}{1.5}$)
		7.79		A1 must be from correct working
				Total 3 marks

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 19*

19	7.5, 8.5, 6.615, 6.625, 1.15, 1.25		3	B1 For a correct upper or lower bound Allow $8.4\dot{9}$ for 8.5, $6.624\dot{9}$ for 6.5, $1.24\dot{9}$ for 1.25
	$(G=)\dfrac{7.5}{2\times6.625-3\times1.15}\left(=\dfrac{7.5}{13.25-3.45}=\dfrac{7.5}{9.8}=\dfrac{75}{98}\right)$			M1 $\dfrac{LB_c}{2\times UB_f - 3\times LB_h}$ where $7.5 \le LB_c < 8$ $6.62 < UB_f \le 6.625$, $1.15 \le LB_h < 1.2$ SCB1 for $\dfrac{7.5}{6.625-1.15}$ $(=1.369(8...))$ [in addition to the first B1]
	working required	0.765		A1 awrt 0.765 dep on completely correct bounds (0.7653061224…) dep on M1
				Total 3 marks

Question 25 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 5*

Question	Working	Answer	Mark	Notes
5 (a)		2.745	1	B1
(b)		2.755	1	B1 allow $2.754\dot{9}$
(c)	$(80 \times 60) \div 2^2$		2	M1 For two of 80, 60, 2 or 4 rather than 2^2 oe
	eg $(80 \times 60) \div 2^2 = 1200$ oe *working with rounded values seen required*	1200		A1 dep on M1 for answer coming from use of the 3 rounded numbers – if 1200 seen then ignore any other working and comments
				Total 4 marks

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